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## Warning: There were 2 warnings in `mutate()`.

## The first warning was:

## i In argument: `clock_min = lubridate::hour(hm(`Booket: Clock Time`)) * 60 +
##   lubridate::minute(hm(`Booket: Clock Time`))`.

## Caused by warning in `.parse_hms()`:

## ! Some strings failed to parse

## i Run `dplyr::last_dplyr_warnings()` to see the 1 remaining warning.

```

Introduction

Research suggests that there is a relationship between hormonal patterns across the span of a day and conditions including sleep disorders and depression. In this study, the focus is on the hormones, DHEA and Cortisol, which play a role in regulating stress. Cortisol is an important steroid hormone that significantly affects metabolism which in turn affects the central nervous system. DHEA is

In this study, researchers followed the cortisol and DHEA patterns over the course of a day for a total study period of three days to measure the patterns of the two hormones in a natural environment.

Researchers utilized two methods to collect saliva samples; a Saliva Procurement and Integrated Testing (SPIT) booklet and an automatic Electronic Monitoring Cap. Saliva samples were collected at waking, 30 minutes after waking, before lunch, and 600 minutes after waking. Subjects did not have to adhere to a specific wake up time for 31 healthy control subjects.

This report addresses the following hypotheses of interest:

Hypothesis 1: There was satisfactory agreement between the subject's recording of sampling times compared to the recording times by the electronic monitoring cap. More specifically, there is a one-to-one relationship between the subject's recorded times and the booklet times.

Hypothesis 2: Subjects adhered appropriately to the study protocol of the +30 minute and +600 minute sampling times.

Hypothesis 3: DHEA and Cortisol levels show an initial increase from waking to 30 minutes followed by a gradual decline from 30 minutes and on.

We will further refer to the electronic monitoring cap as MEM time and the SPIT Booklet as Booklet time for the remainder of the report.

Methods

Preliminary Data Analysis

As missingness can present serious issues for the data analysis if not explored and addressed, we investigated missingness through calculating the number and percentage of missing values for each variable in the dataset. We further visualized missingness for each variable.

Below is a Table1 of the primary variables of interest for the analysis:

For the variables we employed in our analyses, we assessed and visualized distributions to determine if there were any outliers or considerations that needed to be taken for data analysis.

Data Cleaning

Data Analysis

1: To address research question 1, we employed a longitudinal linear mixed effects model with Booklet time in Minutes for the outcome, MEM in Minutes and the day number (1-3) as covariates We included a random intercept for subject that allowed us to take variability between subjects into account in our model. Adding Day as a covariate noticeably improved diagnostic plots for the assumptions of heteroscedasticity and normality. We additionally assessed bias by.... COME BACK TO THIS

2:

2:

Results

From the missingness investigation, we determined the MEMs had the highest amount of missing: 177 missing values which is equivalent to 47.6%. Booklet had the second highest amount of missing with 55 missing values or 14.8%. Cortisol and DHEA measurements had 5 missing values or 1.34% missingness.

The graphic below allows us to visually see the quantity of missingness in the MEM times compared to the other variables in the dataset:

Hypothesis 1

Below is a table of the estimates, 95% Confidence Intervals and P-values for the model:

Booklet Minutes

Predictors

Estimates

CI

p

MEM C

0.99

0.97 – 1.01

<0.001

DAYNUMB2

-3.15

-6.28 – -0.02

0.049

DAYNUMB3

-1.64

-4.79 – 1.51

0.298

N SubjectID

28

Observations

72

Marginal R2 / Conditional R2

0.991 / 0.997

Holding day constant, a one-minute increase in MEM-recorded wake time is associated with an approximately one-minute increase in Booklet-recorded wake time ($\beta \approx 0.99$; 95% CI: (0.97–1.01); $p < 0.0001$). This near one-to-one association indicates strong concordance between the two wake-time measures within a given day. In other words, the subject's recorded wake up time and the MEM's recorded wake up time track each other quite closely. Changes in MEM time are mirrored almost exactly by changes in Booklet time and our initial hypothesis is correct.

Discussion